Online Supplement to "Reproducible Answer Selection over Large Language Model Ensembles, with an Exact Kemeny Reference and a Ground-Truth Audit"

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This supplement collects material that is not central to the argument of the main text: the extended threshold analysis and its proofs, the additional selectors, the coverage tables, the open-ended pilot, and two implementation notes. Section, table and result numbers prefixed with S refer to this supplement.

S1 Extended threshold analysis

This section extends Section 4 of the main text beyond $N = 3$, with pair-dependent judges, correlated wrong answers, a large-N lower bound and the cost term.

Lemma S1 (Monotone coupling)**.** *For every N and every ρ, the probability that the Copeland selector returns a correct candidate is nondecreasing in each q_{ij}.*

Corollary S1 (Pair-dependent judges)**.** *Let $q^*(p, N, \rho)$ be the smallest uniform q at which the selector ties voting. A judge with $q_{ij} \geq q^*$ for all mixed pairs is at least as accurate as voting; one with $q_{ij} \leq q^*$ for all mixed pairs is at most as accurate.*

Table S1 gives q^* computed exactly for $N = 3, \ldots, 7$ (the $N = 3$, $\rho = 0$ column reproduces Table 1). Two things stand out. With distinct wrong answers the threshold *rises* with N: at $p = 0.8$ it goes from 0.804 to 0.935, and Proposition S1 below shows that this is not an artefact of small N: the threshold cannot fall back towards chance as N grows. With correlated wrong answers ($\rho = 0.5$) voting is weaker and the threshold is lower, about 0.82 at $p = 0.8$ for $N \geq 4$. The direction of the relaxation is therefore the one a reader would expect, but its size is large enough that the realistic threshold is far from $\frac{1}{2}$. For GPQA Diamond, whose wrong answers are heavily correlated, this lowers the threshold from 0.787 to 0.714 (Table 8); the measured $\hat{q} = 0.536$ remains far below it. We checked the exact enumeration against an independent Monte Carlo simulation with common random numbers (Table S2; all four configurations agree within sampling error) and checked Corollary S1 on random pair-dependent judges: in 30 of 30 random $N = 5$ instances the accuracy gap lies between the gaps at $q_{\min}$ and $q_{\max}$.

Why the threshold keeps rising: a large-N lower bound With distinct wrong answers plurality's error has a closed form. It is correct whenever at least two candidates are correct, and when exactly one is correct all N answers differ and it is chosen with probability $1/N$, so

$$\mathrm{err}_V(N) = (1-p)^N + (N-1)\, p\, (1-p)^{N-1}, \tag{S1}$$

Table S1: Exact break-even pairwise accuracy $q^*(p, N, \rho)$ for the Copeland selector against plurality voting, with per-candidate accuracy p and wrong-answer correlation ρ.

	$\rho = 0$		$\rho = 0.5$	
N	$p = 0.5$	$p = 0.8$	$p = 0.5$	$p = 0.8$
3	0.667	0.804	0.625	0.782
4	0.759	0.891	0.675	0.814
5	0.800	0.919	0.682	0.824
6	0.825	0.935	0.679	0.819
7	0.841	0.945	0.673	0.821

Table S2: Exact accuracy gap (Copeland minus plurality, percentage points) against a 200,000-draw Monte Carlo estimate with its standard error.

N	p	ρ	q	Exact	Monte Carlo
4	0.5	0.0	0.7	-5.40	-5.50 ± 0.11
5	0.8	0.5	0.8	-0.66	-0.68 ± 0.05
6	0.5	0.5	0.7	+2.64	+2.65 ± 0.11
6	0.8	0.0	0.9	-0.20	-0.20 ± 0.01

which decays at rate $-\ln(1-p)$ per candidate. A pairwise-Copeland selector that errs on the event "a wrong candidate beats a sizeable block of correct ones" decays more slowly unless q is large. Write $\mathrm{KL}(\eta\|p) = \eta \ln \frac{\eta}{p} + (1-\eta) \ln \frac{1-\eta}{1-p}$ and

$$\Lambda(p, q) \;=\; \inf_{\eta > 1 - \frac{1}{2q}} \Big[\mathrm{KL}(\eta\|p) + \eta \ln \tfrac{1}{1-q} \Big].$$

Proposition S1 (Large-N lower bound)**.** *Let wrong answers be distinct ($\rho = 0$) and the judge uniform with $q > \frac{1}{2}$. If $\Lambda(p, q) < -\ln(1-p)$, there is N_0 such that for all $N \geq N_0$ the Copeland selector is strictly less accurate than plurality. Consequently* $\liminf_{N\to\infty} q^*(p, N) \geq q_\Lambda(p)$*, where $q_\Lambda(p)$ is the unique root of $\Lambda(p, q) = -\ln(1-p)$.*

Table S3 lists $q_\Lambda(p)$ next to the exact $N = 6$ thresholds: the exact values sit just below the limiting lower bound for $p \geq 0.6$ and just above it at $p = 0.5$, as a sequence rising to (at least) q_Λ would. We do not prove that the limit equals q_Λ; the proposition is a lower bound, obtained by exhibiting one specific way in which Copeland errs.

Table S3: Plurality's error decay rate $-\ln(1-p)$, the large-N lower bound $q_\Lambda(p)$ on $\liminf_N q^*(p, N)$ (Proposition S1), and the exact $q^*(p, 6)$ ($\rho = 0$).

p	$-\ln(1-p)$	$q_\Lambda(p)$	$q^*(p, 6)$
0.5	0.693	0.820	0.825
0.6	0.916	0.867	0.862
0.7	1.204	0.907	0.899
0.8	1.609	0.943	0.935
0.9	2.303	0.973	0.970

Structure of the wrong answers beyond a single attractor The attractor model with parameter ρ is one way to correlate wrong answers. We also drew each wrong answer uniformly

from D wrong values ($D = 1$: all wrong answers coincide; $D = 3$: the three distractors of a four-option question; $D = \infty$: all distinct). Table S4 shows that the threshold is monotone in the structure: the fewer distinct wrong values, the weaker plurality and the lower the threshold, down to $q^* = \frac{1}{2}$ when all wrong answers coincide and the pool is small ($N = 4$, $p = 0.5$). For the four-option GPQA questions ($D = 3$) the threshold is 0.64–0.80 at $N = 4, 5$, consistent with the attractor-model estimate for GPQA in Table 8.

Table S4: Exact q^* when each wrong answer is uniform over D wrong values.

	$N = 4$		$N = 5$	
D	$p = 0.5$	$p = 0.8$	$p = 0.5$	$p = 0.8$
1	0.500	0.669	0.500	0.721
2	0.594	0.754	0.592	0.770
3	0.640	0.791	0.638	0.798
8	0.710	0.847	0.723	0.854
∞	0.759	0.891	0.800	0.919

The cost term The results above use the quality term alone, the case most favourable to the judge. If the cost (length) of a candidate is independent of its correctness, adding it can only move the selector towards the shortest-candidate rule, whose accuracy is exactly p, the accuracy of a random pick; Table S5 shows by simulation how accuracy falls from the judge-only value to p as the cost weight λ (relative to the Copeland score scaled to $[0, 1]$) grows. In our data lengths are not independent of correctness (shortest-candidate accuracy differs from the random-pick accuracy, Table S6), so the empirical effect of the cost term is a property of the data, which Section 6.5 measures directly.

Table S5: Simulated accuracy (%) of selection by Copeland score minus $\lambda\times$ an independent random length, $N = 3$, $p = 0.8$ (plurality voting: 92.8%; random pick: 80.0%), 100,000 draws.

λ	$q = 0.6$	$q = 0.9$
0	84.5	96.1
0.25	84.4	96.1
1	83.9	94.9
4	81.4	86.0
∞ (shortest)	79.8	79.8

Large N: a regime split, not a threshold For large N the break-even accuracy is no longer informative, because both selectors approach perfect accuracy whenever correct answers form a plurality; the difference then lives in exponentially small error rates, which a threshold at fixed accuracy does not capture. What can be said rigorously is a consistency result.

Proposition S2 (Large-N regimes)**.** *Let candidates be independent with correct-answer probability $p > 0$, wrong answers taking finitely many values, and let m be the largest probability of any single wrong answer. (i) Plurality voting is correct with probability tending to 1 if $p > m$ and to 0 if $p < m$. (ii) If $q_{ij} \geq q_{\min} > \frac{1}{2}$ for all mixed pairs, the Copeland selector is correct with probability at least $1 - (1 - p)^N - 2N^2 e^{-(q_{\min} - 1/2)^2 N/2} \to 1$, whatever m.*

So a judge that is better than chance can rescue a generator pool in which the single most popular answer is wrong ($p < m$), where plurality voting fails with certainty, but it cannot be expected to improve a pool in which the right answer already leads. This is the pairwise-judge analogue of the dichotomy established for reward-model best-of-N and majority voting by Di et al. (2025), and we claim consistency with it, not novelty: the contribution here is the exact finite-N thresholds above and their confrontation with the measured $\hat{q}$. Our benchmark pools are in the right-majority regime (the shipped Nemotron generator is correct on 77–99% of candidates, floor column of Table 5), which is why the judge cannot help.

S2 Proofs for the extended threshold analysis

Lemma S1. Fix a mixed pair (c, w) with c correct and w wrong and condition on all other comparison outcomes; the outcome of (c, w) is a Bernoulli(q_{cw}) variable that is the only one depending on q_{cw}, so it suffices to show that changing it from "w wins" to "c wins" cannot lower the probability that the selected candidate is correct. The change raises c's score by one and lowers w's by one. Let T be the set of top scorers before and T' after, and let the selected candidate be uniform on the top set, so the success probability is the share of correct candidates in it. If $c \in T$, then $T' = \{c\}$, which is correct. If $c \notin T$ and $w \notin T$, then $T' = T$ or $T' = T \cup \{c\}$; adding a correct candidate cannot lower a share that is at most one. If $w \in T$ and $|T| > 1$, then T' removes w and possibly adds c, which cannot lower the share. If w is the unique top scorer, the share before is 0. In every case the share does not decrease. Averaging over the other outcomes gives the claim, and Corollary S1 follows by comparing with the uniform judges at $q_{\min}$ and $q_{\max}$ (voting does not depend on the judge).

Proposition S2. (i) The count of the correct answer is Binomial(N, p) and that of each wrong answer is Binomial$(N, \cdot)$ with success probability at most m, the largest attaining m; by Hoeffding's inequality and a union bound over the finitely many answers, the correct answer is the unique plurality with probability tending to one if $p > m$, and is beaten by the most popular wrong answer with probability tending to one if $p < m$. (ii) Condition on the set of correct candidates, of size $k_c \geq 1$ (probability $1 - (1-p)^N$), and let $k_w = N - k_c$. A correct candidate's score is a sum of $N - 1$ independent comparisons with mean at least $q_{\min} k_w + \frac{1}{2}(k_c - 1)$; a wrong candidate's has mean at most $(1 - q_{\min})k_c + \frac{1}{2}(k_w - 1)$; the difference is at least $(q_{\min} - \frac{1}{2})N$. By Hoeffding's inequality each score is within $t = (q_{\min} - \frac{1}{2})N/2$ of its mean with probability at least $1 - 2\exp(-2t^2/(N-1)) \geq 1 - 2\exp(-(q_{\min} - \frac{1}{2})^2 N/2)$. A union bound over the N candidates gives the stated probability that every correct candidate outscores every wrong one, in which case the selected candidate is correct. □

Equation (S1). With distinct wrong answers the plurality is the correct answer exactly when at least two candidates are correct (its count then exceeds every wrong count, which is one); with one correct candidate all N answers are tied and the correct one is chosen with probability $1/N$; with none it cannot be chosen. Hence $\mathrm{err}_V = (1-p)^N + Np(1-p)^{N-1}(1 - 1/N)$.

Proposition S1. Fix $q > \frac{1}{2}$ and $\eta \in (1 - \frac{1}{2q}, 1)$ and let $K = \lceil \eta N \rceil$. Consider the event E that exactly K candidates are correct and a designated wrong candidate w beats all of them (probability $\binom{N}{K} p^K (1-p)^{N-K} (1-q)^K \geq \frac{1}{N+1} e^{-N\,\mathrm{KL}(K/N \| p)} (1-q)^K$). Conditionally on E, w's score is $K + \mathrm{Bin}(N - K - 1, \frac{1}{2})$, with mean $(N + K - 1)/2$; a correct candidate's score has mean at most $q(N - K - 1) + (K - 1)/2 + 0$ (it has already lost to w), and another wrong candidate's has mean $(1 - q)K + (N - K - 1)/2$. The gaps are $N(\frac{1}{2} - q(1 - \eta)) + O(1)$ and qK, linear in N and positive because $\eta > 1 - \frac{1}{2q}$. By Hoeffding's inequality and a union bound over the N candidates, w is the unique top scorer with conditional probability $1 - O(Ne^{-cN})$, so the Copeland selector

errs on E with probability at least $\frac{1}{2}$ for large N. Therefore $\mathrm{err}_C \geq \frac{1}{2(N+1)} e^{-N[\mathrm{KL}(\eta\|p)+\eta \ln \frac{1}{1-q}+o(1)]}$. If $\Lambda(p,q) < -\ln(1-p)$ choose η with bracket below $-\ln(1-p)$; since $\mathrm{err}_V \leq N(1-p)^{N-1}$ by Equation (S1), $\mathrm{err}_C > \mathrm{err}_V$ for all large N. For the consequence: $\Lambda(p,q)$ is continuous and strictly increasing in q (it equals an infimum of functions increasing in q over a set that shrinks with q), lies below $-\ln(1-p)$ for q slightly above $\frac{1}{2}$ (the derivative of $\mathrm{KL}(\eta\|p)$ is $-\infty$ at $\eta = 0$) and diverges as $q \uparrow 1$ (it is at least $\frac{1}{2}\ln\frac{1}{1-q}$), so $q_\Lambda(p)$ exists and is unique. For any $\epsilon > 0$ the accuracy gap at $q_\Lambda - \epsilon$ is negative for large N, and by Lemma S1 it is nonincreasing as q decreases, so it is negative for every $q \leq q_\Lambda - \epsilon$. Hence $q^*(p,N) > q_\Lambda - \epsilon$ for large N. □

S3 Additional selectors

Table S6 compares OR-Select's competitors beyond Section 6.2. Judge-free selectors (shortest and longest candidate, and minimum-Bayes-risk selection with token-set Jaccard similarity) run on every graded problem; judge-based selectors are replayed on the problems whose judge outputs were stored (pointwise confidence argmax, a vote weighted by the scorer's confidence, a knockout tournament over the pairwise matrix, and a Bradley–Terry fit). Each is paired with self-consistency on the same problems; the parenthesis in each accuracy cell is the self-consistency accuracy on those problems.

No selector beats plurality voting in either condition. The judge-free minimum-Bayes-risk selector, which only measures how much a candidate resembles the others, stays within a few problems of voting in both conditions, and so does the confidence-weighted vote, which is itself a variant of voting. The selectors that act on the judge's pairwise or pointwise output choose worse than voting in heterogeneous pools, as OR-Select does, and the trivial length rules (shortest, longest) are worse still. The common thread is the same as in Section 6.5: whatever signal the judge adds is not about correctness.

Table S6: Additional selectors, paired with self-consistency (SC) on the same graded problems. For each condition: problems n, accuracy % (SC accuracy % on the same problems), problems where the selector is right and SC wrong over the reverse, and the exact McNemar p. †: needs the stored judge outputs, so n is the replayable subset.

	Homogeneous				Heterogeneous			
Selector	n	Acc. (SC)	W/L	p	n	Acc. (SC)	W/L	p
Shortest candidate	532	84.6 (85.5)	10/15	0.424	402	78.4 (85.1)	11/38	<0.001
Longest candidate	532	83.3 (85.5)	11/23	0.058	402	80.1 (85.1)	5/25	<0.001
MBR (Jaccard)	532	85.9 (85.5)	9/7	0.804	402	85.3 (85.1)	9/8	1.000
Confidence argmax†	332	80.7 (80.7)	7/7	1.000	397	79.8 (84.9)	10/30	0.002
Confidence-weighted vote†	332	79.8 (80.7)	0/3	0.250	397	84.4 (84.9)	3/5	0.727
Knockout tournament†	332	79.8 (80.7)	11/14	0.690	397	79.6 (84.9)	12/33	0.002
Bradley–Terry†	332	80.1 (80.7)	10/12	0.832	397	77.8 (84.9)	8/36	<0.001

S4 Coverage and higher-cap second pass: tables

Table S7: Coverage by track. Truncated: no valid answer after resampling because generation hit the output cap. Other: failed for another reason. Untried: not attempted. With Laguna: graded heterogeneous problems that still contain Laguna-S candidates.

Track	Total	Graded	Truncated	Other	Untried	With Laguna
B — GSM8K, homog.	40	40	0	0	0	–
B′ — GSM8K, heterog.	40	40	0	0	0	40
C — MATH L5, homog.	134	134	0	0	0	–
C′ — MATH L5, heterog.	134	134	0	0	0	131
D — AIME 2026, homog.	30	30	0	0	0	–
D′ — AIME 2026, heterog.	30	30	0	0	0	23
E — GPQA Diamond, homog.	198	198	0	0	0	–
E′ — GPQA Diamond, heterog.	198	198	0	0	0	80
F — AIME 2025, homog.	30	30	0	0	0	–
G — MMLU-Pro, homog.	100	100	0	0	0	–

Table S8: Effect of the flagged second pass. Main protocol: problems that completed under the 32,768-token cap. Rescued: problems completed only under the higher cap. Final: all graded problems.

Track	Main n	Main OR	Main SC	Rescued	Final OR	Final SC
C — MATH L5, homog.	132	90.9	92.4	2	89.6	91.0
C′ — MATH L5, heterog.	131	79.4	90.8	3	79.1	90.3
D — AIME 2026, homog.	24	95.8	91.7	6	90.0	86.7
D′ — AIME 2026, heterog.	23	78.3	100.0	7	63.3	93.3
E — GPQA Diamond, homog.	175	82.3	82.9	23	78.8	78.8
E′ — GPQA Diamond, heterog.	163	85.3	82.8	35	79.3	77.3
F — AIME 2025, homog.	24	87.5	95.8	6	76.7	83.3
G — MMLU-Pro, homog.	94	88.3	88.3	6	86.0	86.0

S5 Pilot illustration: open-ended queries

Three open-ended queries ($N = 5$ each, no ground truth) were run in an earlier pilot to check that the pipeline behaves sensibly where self-consistency has nothing to vote on. Across the 15 candidates, none was byte-identical to another, so self-consistency resorted to an arbitrary tie-break on every query, and OR-Select differed from both baselines on all three (score margins 0.088, 0.446 and a near-tie of 0.003). With three queries and no ground truth this is an illustration, not evidence about quality. Two pilot ablations on the same candidate sets (setting $w_4 = 0$; forward-only judging) changed 0 and 2 of the 3 winners respectively: the self-consistency term was inert because no candidate set clustered, and bidirectional judging mattered in two of three cases, consistent with position bias being a first-order judge failure mode (Zheng et al., 2023; Tang et al., 2024; Wang et al., 2023). Three instances cannot characterize when the directions disagree.

S6 A concrete non-determinism finding

In an earlier pilot run on a different provider and model, three candidates had byte-identical text. By construction every pairwise weight among identical text must be 0.5, since the forward and backward judge calls are the same prompt and $(v + (1 - v))/2 = 0.5$. The observed quality scores were 0.375, 0.50 and 0.625, direct evidence that the provider's `seed` parameter was not reliably honored in that run, in line with systematic run-to-run variation reported for nominally deterministic settings (Atil et al., 2024). This is a single observation; the exact-match short-circuit (Section 3.1) removes this failure mode for identical text but not for different texts that receive unstable scores, which remains open.

S7 Retry independence and extraction fallbacks

Each validity-retry draws with a freshly varied seed, so a persistently invalid slot gets three independent samples rather than three repeats of what a seed-honoring backend already returned. Extraction accepts more than one marker: GPQA also recognizes a prose "the answer is X" pattern alongside `ANSWER: X` and `\boxed{X}`, because its answer space is four letters; MATH's open-ended `\boxed{}` extraction has no prose fallback, since a wrong guess there would convert an honest "ungraded" problem into a confidently incorrect one. This asymmetry is a design choice.

S8 Worked example and positional-bias figure

Table S9 and Figure S1 decompose Equation (1) for one real open-ended run ($N = 5$, no exact-text duplicates). quality_i is recovered by inverting the saved score_i against the known confidence, cost and consistency, and the recovered argmax is checked by script against the saved winner.

Table S9: Component scores for one real open-ended run (winner marked *). quality_i is recovered by inversion, not stored directly.

Candidate	quality_i	confidence_i	consistency_i	cost_i	score_i
c0	0.606	1.00	0.20	38	0.486
c1	0.387	1.00	0.20	32	0.327
c2*	0.644	1.00	0.20	33	0.574
c3	0.575	1.00	0.20	38	0.455
c4	0.288	1.00	0.20	33	0.218

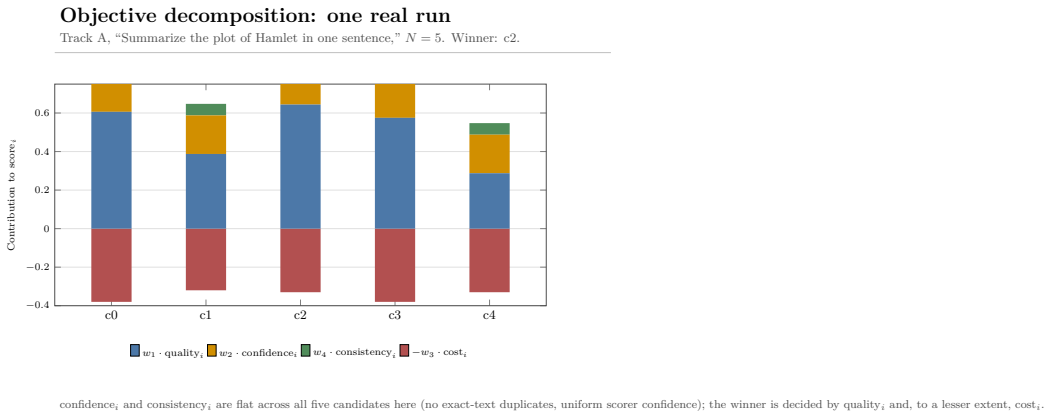


Figure S1: The same run as Table S9, as stacked weighted contributions to score_i. Confidence and consistency are constant across candidates here (no duplicates; uniform scorer confidence), so the ranking is decided by quality and cost. These candidates are one-sentence summaries; on multi-hundred-word reasoning outputs the cost term is far larger relative to quality (Section 6.5).

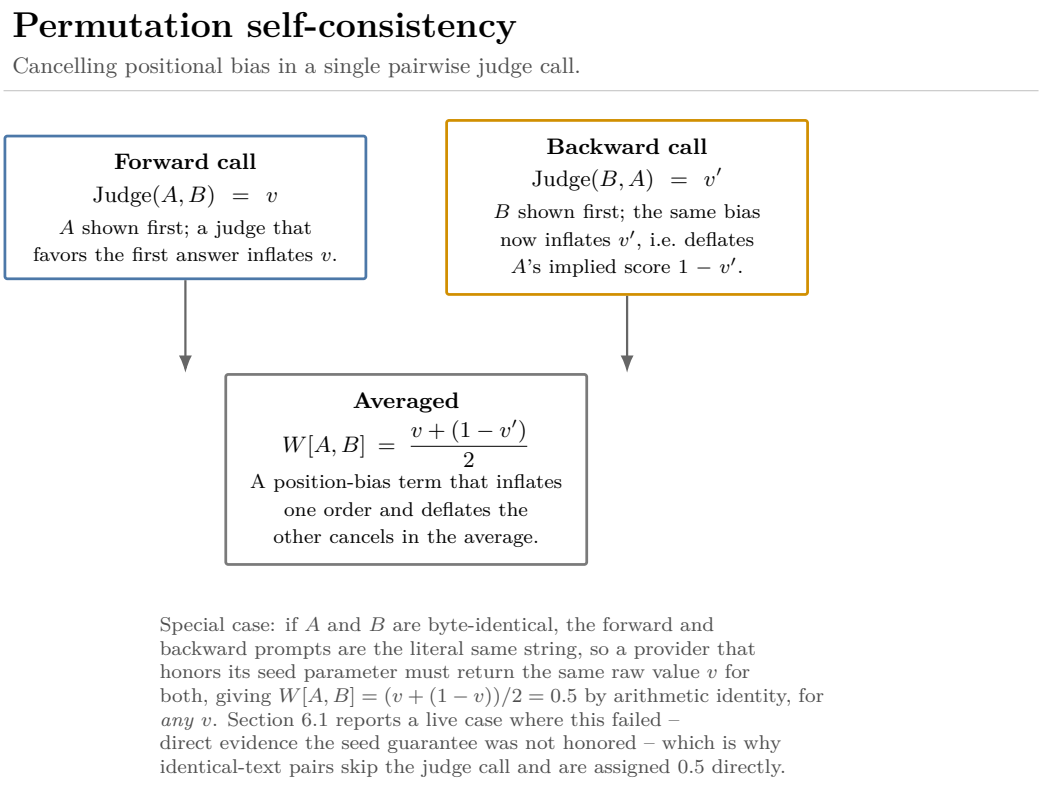


Figure S2: Permutation self-consistency cancels positional bias by averaging the forward and (complement-transformed) backward judge calls for each pair. Byte-identical candidates force $W = 0.5$ by arithmetic identity, independent of the judge's output.

References

Atil, B., Aykent, S., Chittams, A., Fu, L., Passonneau, R. J., Radcliffe, E., Rajagopal, G. R., Sloan, A., Tudrej, T., Ture, F., Wu, Z., Xu, L., & Baldwin, B. (2024). *Non-determinism of "deterministic" LLM settings*.

Di, Q., Ji, K., Li, X., Zhao, H., & Gu, Q. (2025). *Best-of-majority: Minimax-optimal strategy for pass@k inference scaling*. ICLR 2026.

Tang, R., Zhang, X., Ma, X., Lin, J., & Ture, F. (2024). Found in the middle: Permutation self-consistency improves listwise ranking in large language models. *Findings of the Association for Computational Linguistics: NAACL 2024*.

Wang, P., Li, L., Chen, L., Cai, Z., Zhu, D., Lin, B., Cao, Y., Liu, Q., Liu, T., & Sui, Z. (2023). *Large language models are not fair evaluators*.

Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E. P., Zhang, H., Gonzalez, J. E., & Stoica, I. (2023). Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in Neural Information Processing Systems (NeurIPS)*.